

Portfolio Case Studies: Work That Speaks for Itself

Voice Card

Direct, technically grounded, precise, clear, no hype.

Portfolio Bio & Contact / Call to Action

Bio

I build production-grade Python backends, AI agents, and async systems that turn heavy data workflows into simple API responses. My focus is on architecture that performs under pressure—clean routing, vector search pipelines, and reliable cloud deployments.

Contact / CTA

Looking for a backend engineer to build reliable AI infrastructure or async APIs? Let's talk architecture. Reach out at muhammadhamzanawaz@gmail.com.

Case Studies

1. NeoRetail Automation Hub — AI-Powered Retail Operations Platform

- **The Problem:** Small to mid-sized retail operators struggle to integrate customer acquisition, AI engagement, and database tracking without piecing together fragmented, expensive SaaS tools.
- **What I Did & Decided:**
 - Architected a unified multi-module AI backend using FastAPI and Gemini API to handle lead routing, CRM functions, and automated marketing workflows.
 - Configured a hybrid PostgreSQL storage setup with vector extensions for context retrieval.
 - Containerized the stack via Docker and configured Nginx reverse proxies for staging, leading to a single-click deployment pipeline on Railway with public environment configurations.

- **What Came of It:** Delivered a fully operational, live retail automation system running on cloud infrastructure, reducing API query latencies for automated customer interactions.

2. HireFlow — Vector-Search Recruitment Platform

- **The Problem:** Standard keyword matching on resumes fails to capture semantic relevance, leaving recruiters drowning in candidate files while missing high-fit applicants.
- **What I Did & Decided:**
 - Built an automated recruitment engine powered by FastAPI, integrating ChromaDB for high-dimensional vector embeddings and similarity search.
 - Implemented WebSockets to stream real-time parsing status and match notifications directly to the recruiter's interface.
 - Designed modular async pipelines to ingest, parse, and score incoming resume files concurrently without locking thread pools.
- **What Came of It:** Reduced candidate search times from manual file skimming to instant vector query results, surfacing contextually matched profiles across thousands of stored applications.

3. Full-Stack FastAPI Engine — Auth & Database Migration Architecture

- **The Problem:** Scaling web applications requires a robust database schema and secure session handling that won't break during future feature updates or high concurrent user load.
- **What I Did & Decided:**
 - Engineered a full-stack FastAPI application featuring structured JWT authentication and hashed credential storage.
 - Integrated PostgreSQL with SQLAlchemy ORM and Alembic for declarative data models and controlled schema migration tracks.

- Enforced clean separation of concerns across models, schemas, and API routers to prevent tight coupling as the codebase scales.
- **What Came of It:** Built a production-ready template that eliminates schema drift, allowing instant database rollbacks and secure route protection across backend endpoints.

Before & After: Generic AI vs. Edited Voice

- **Generic AI Line:** *"Leveraging cutting-edge multi-agent orchestration frameworks and state-of-the-art vector embedding solutions, this innovative platform revolutionizes how modern enterprises streamline complex recruitment workflows."*
- **Edited Version:** *"I built an async FastAPI pipeline backed by ChromaDB vector search to instantly rank resumes by actual technical fit instead of simple keyword matches."*